

ML Assignment: Aerial Cell Tower Antenna Detection

Technical and Performance Report

Submitted by: Sarah Saju Muhammed

B tech-Electronics and Communication Engineering

Amrita Vishwa Vidyapeetham

13 April 2025

1. Objective

Train and deploy an object detection model that identifies only the antenna portions of cell towers from aerial video footage ignoring the rest of the metal framework.

2. Dataset Summary

Source: A collection of aerial cell tower images sourced from Google Drive through open internet search results. There are a total of 1000 images out of which 261 images were annotated for training and validation purposes (65%training & 35% testing).The dataset contains images of different parts of the tower such as the top nest , bottom nest and middle nest , and the images having high sun exposure were also included.

Image Resolution: The original dataset contains high resolution images with 5472 x 3648 pixels. No manual resizing or preprocessing was done as the YOLOv8 framework I used automatically resizes inputs to 640 x640 during training.

Classes and Labelling : The dataset contains only one class ie) Antenna. Each annotated image contains bounding boxes that mark the antenna portions of the cell towers. The annotations were done in YOLO format.

Negative Samples: To improve the model's accuracy and prevent false positives , images of tower parts that do not contain antenna and white cylindrical objects were added as negative samples. These images were used to represent non-antenna structures and were left unlabelled.

Annotation Format: Tool used :CVAT.Bounding boxes for segmentation. Saved in YOLO format(.txt) ,(class , class x_centre , class y_centre , width of bounding box , height of bounding box).

3. Model and Framework Used

Model Chosen: YOLOv8s

Framework: Ultralytics YOLOv8 (Pytorch based)

Reason for choice: Lightweight for drone deployment .Accurate in detecting smaller objects and faster compared to YOLOv8m.

4. Training Strategy and Hyperparameters

- **Training Set Size:** 64.75%(169 images)
- **Validation Set Size:** 35.24%(92 images)
- **Epochs:**100
- **Batch Size:**16
- **Image size:**640
- **Optimizer:** AdamW(lr=0.002, momentum=0.9) with parameter groups 77
weight(decay=0.0)-77
weight(decay=0.0005)- 83
bias(decay=0.0)-83
- **Image size:** 640

- **Learning Rate:**0.01
- **Data Augmentation :**Enabled (default YOLOV8 settings).
Mosaic Augmentation
Horizontal flip
HSV colour Augmentation
Scaling and Translation.
- **Early Stopping:** enabled but was not triggered and the model completed all 100 epochs of training
- **Model Checkpointing:** Enabled.YOLOv8 automatically saves both best.pt (best performing model) and last.pt(final model).last.pt was used for inference

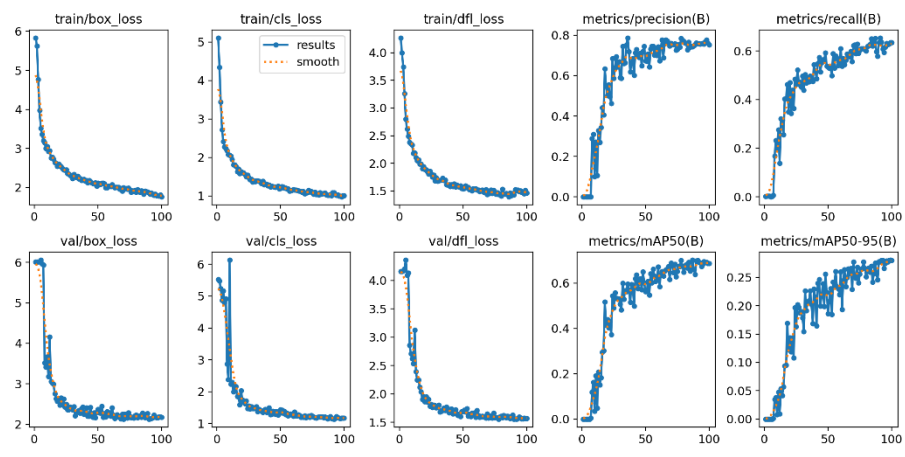
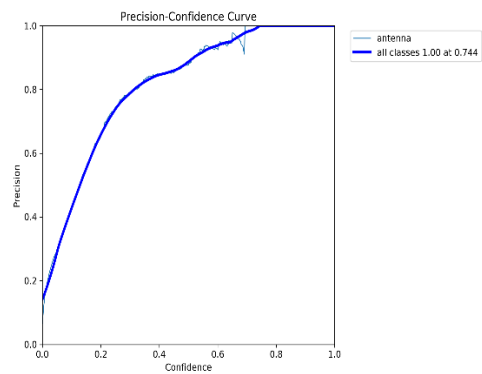
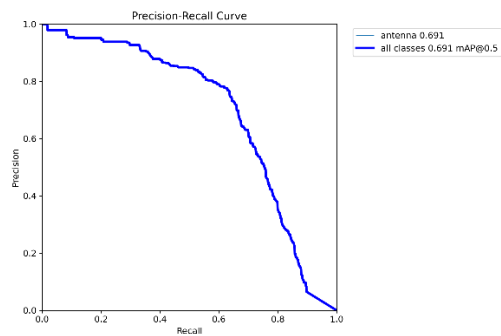
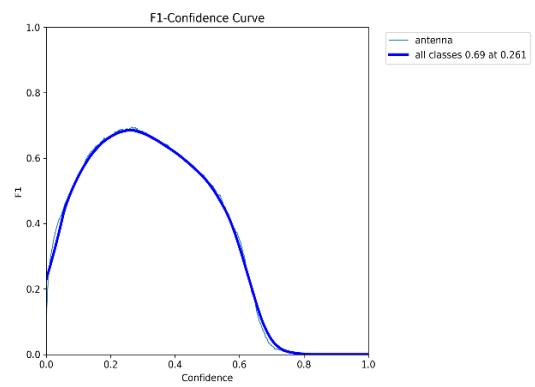
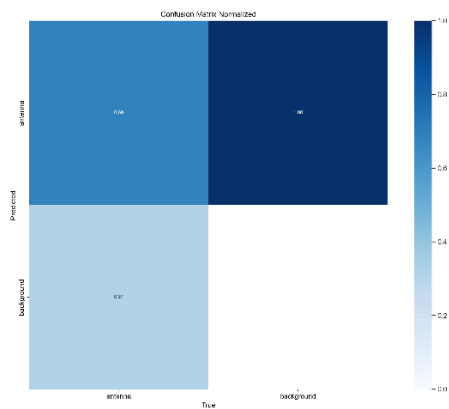
5. Inference Pipeline and Optimizations

Inference video source: Drone Inspection Video-YouTube

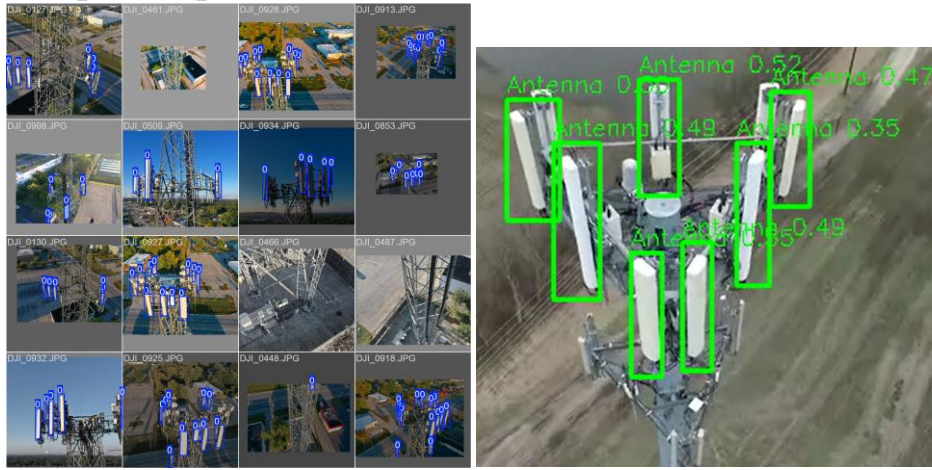
1. Video converted to frames using opencv
2. **Model Input:** Frame resized to 640x640 by default and normalised automatically by YOLOv8 and slowed by factor 2 to improve readability
3. **Output:** Bounding Boxes over antenna portions only with confidence threshold >0.3 .
4. Manually annotated the inference video to compare and compute the Hit Ratio, Bounding Box Accuracy etc
5. **Post-Processing-** Non-max suppression(NMS),class filtering(Single Class-Antenna)
6. **Output Storage:** Antenna detected video was saved.Bounding boxes coordinates of antennas were recorded in a csv file

6. Performance Evaluation

Metric	Value
Precision	77.63%(98 th epoch)
Recall	63.402%
mAP@0.5	70.034%(91 st epoch)
Inference FPS	2.5177
Model Size	42.9 MB
Inference Latency	0.397s
Bounding Box Accuracy	68.6%
Hit Ratio	63.4%
Memory Usage	150MB



Sample Output



7. Conclusion

In this project, an object detection model was developed and successfully deployed to identify the antenna portions of cell towers from aerial video footage. The model achieved 77.63% precision and 63.402% recall rates, demonstrating its effectiveness in detecting antennas while ignoring the metal framework. Memory usage during inference averaged around 150MB, making the model lightweight and suitable for deployment on resource-constrained edge devices like drones. Future improvements could include enhancing the dataset with more varied images and experimenting with different model architectures to boost precision further. The system shows promise for practical applications such as drone-based tower inspections, with potential for expanding to other types of aerial object detection.

